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To the Editor-in-Chief,
Computational Economics

Dear Editors,

I am submitting the manuscript “Fair and Transferable Graph Reinforcement Learning for Urban Rail Transit Network Expansion Across Asian Cities” for consideration as a regular research article in *Computational Economics*.

The underlying decision problem is an economic one before it is an engineering one: a city choosing where to extend its rail network is making a large, essentially irreversible public-investment allocation under a fixed station/budget constraint, and must set a trade-off between aggregate transportation-demand efficiency and equitable access to new infrastructure across neighborhoods that differ in income and existing service. That trade-off is normally fixed once, by policy, before a network is built. We formulate it instead as a sequential allocation problem solved by reinforcement learning over a graph-structured representation of the city’s regions and travel demand, with a Tchebycheff-scalarized vector objective (demand satisfaction, social equity, radiation accessibility) that lets a single trained policy be queried at different points on the efficiency-equity frontier rather than committing to one weighting in advance – an agent-based, machine-learning approach to a resource-allocation and welfare trade-off problem, which is the class of question this journal’s scope names directly.

The paper’s contribution is threefold: a graph neural network encoder over the city’s spatial-contiguity and origin-destination relations, feeding a multi-objective actor-critic policy that extends prior single-objective (MetroGNN) and non-GNN multi-objective (Zhang et al. 2024) formulations of the metro network expansion problem; a cross-city curriculum-transfer mechanism, motivated directly by a recent critique (Michailidis et al. 2026) that deep RL’s training cost is unjustified at this problem’s scale, that pretrains on one city’s network and warm-starts another’s to test whether transfer buys back that cost in sample efficiency; and, to our knowledge, the first evaluation of this problem class spanning both a small, data-sparse emerging network (Ho Chi Minh City) and a mature, previously-studied one (Xi’an) in one consistent protocol.

The empirical results are reported as found rather than smoothed toward the hoped-for conclusion. A five-seed sweep on Xi’an finds no reliable advantage for the full method over either architectural ablation on demand, equity, or coverage, and the tabular RL baseline of Michailidis et al. still wins on raw demand under a matched training budget – a negative result for model complexity that we think is itself of interest to a computational-economics readership asking when a more expressive (and more expensive) representation is worth its cost. Tracing an earlier, more favorable-looking equity finding, we discovered GPU training non-determinism severe enough to invert which architecture a metric appears to favor, traced it to two distinct causes, fixed both, and verified deterministic re-runs – a reproducibility hazard we believe generalizes beyond our own implementation to other GNN-based methods built on the same default scatter-reduction operations. On Ho Chi Minh City, an encoder ablation collapses to near-zero demand where the full method does not, and warm-starting from Xi’an’s trained policy shows genuine positive transfer at initialization and, under a reduced training budget, reaches non-trivial demand and consistently higher coverage than training from scratch – direct evidence for the curriculum’s sample-efficiency claim on the one city where we could test it. A separately implemented radiation-model accessibility measure does not share the coverage proxy’s architectural sensitivity,

a construct-validity finding we report rather than omit.

All code – the GNN encoder, multi-objective policy, training and evaluation scripts for both cities, and the full test suite – is publicly available now, not embargoed until publication, at https://github.com/vinhqdang/metro_expansion. Data sources are public third-party releases (Michailidis et al.’s `mo-tndp`, the CityScope/MIT `CSL_HCMC` repository, and `GZUPA/subway-traffic-data-set`), cited and documented in the manuscript’s Data Availability statement.

The manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. I am the sole author and approve its submission. I declare no competing financial or personal interests, and the work received no specific grant from any funding agency. In accordance with the journal’s policy, the manuscript will include a declaration of generative-AI use in its preparation.

Thank you for your consideration.

Yours sincerely,

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